

PARAMPARA

Feel the Master's Hand

One sensor sleeve, from the fingers to the shoulder. Three living crafts: tabla, handloom and puppetry. We measure how a master moves, prove it is the master's, teach it, and step back until the learner performs alone.

Final solution document · version 6

AICTE · MIC Student Innovation · Heritage & Culture · Hardware

4 October 2026

In one minute

- **One innovation:** a master's Skill Fingerprint (how every finger, the wrist, elbow and shoulder move, plus what the tool does), used as a teaching reference.
- **Tabla is the flagship;** handloom and Kathputli puppetry show the same sleeve and code generalise.
- **Working today:** the teaching loop runs on a phone (end-to-end tests pass) and the sleeve firmware core passes 27 unit tests.
- **Exact equipment:** CAD-drawn sleeve with datasheet-checked parts, budgets and eight defined bench tests.
- **Pre-tested on real performers:** performer information survives a new session and style; timing alone is close to our pass mark, so we add body motion.
- **Learning counted with the device off,** and the master owns the recording.

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Citations like [12] point to Section 17. Every reference carries a status: **V** verified by live web search in October 2026, **P** existence verified with some details from memory, **S** a standard textbook or classic paper. Numbers we have not yet measured are marked **[Target]**; engineering estimates are marked **[Estimate]**. All equipment images are **CAD renders of our design, not photographs**; the only screenshot is of software that runs today (Section 5).

1. The idea on one page

The problem

In tabla, handloom weaving and string puppetry, a master's skill lives in the hands and arms: which finger moves, how the wrist, elbow and shoulder work together, and the rhythm of it all. It passes on only face to face, over years. India's handloom workforce fell from 43.32 lakh to 35.22 lakh between the last two censuses [66]; Delhi's puppeteer colony was demolished in 2017 [73]; and archives keep how masters **sound and look** [50], not how they **move**.

Our one innovation

A master's Skill Fingerprint: a measured, consented record of how one master moves (every finger, the wrist, elbow and shoulder) and what their tool does, proven to belong to the master and not to their instrument or the day, and used as a teaching reference.

Tabla is the flagship (about 80% of our effort): its complete validation pathway is defined and the method is pre-tested on real performer data. Handloom and puppetry (about 15%) are **generalisation demonstrations** of the same sleeve and code; the last 5% is the platform roadmap.

The 90 seconds we are building towards

A judge who has never played tabla puts on the sleeve. A master's recorded Teentaal starts. Small pulses on the right fingers and the wrist arrive a moment before each stroke, and the master's own timing plays in the ear. Every few cycles the pulses thin out. Then they stop, the screen goes quiet, and the judge plays sixteen beats alone. A number appears: the unaided score. **That number, earned with the device off, is the whole point of PARAMPARA.**

The teaching half of this moment already runs on a phone today (Section 5); the sleeve adds the fingers, wrist, elbow and shoulder.

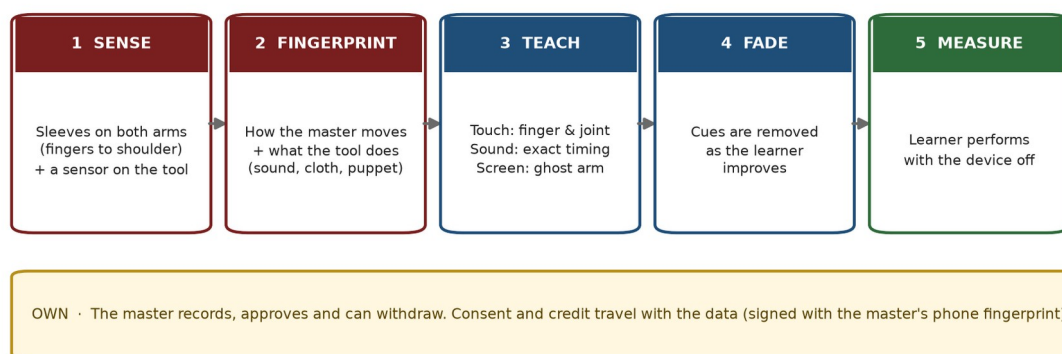


Figure 1. Five steps, plus ownership by the master.

Why three crafts, and why these three

Craft	Kind of skill	What the tool sensor measures	The 'cycle' we compare
Tabla (flagship)	Fast striking rhythm	A piezo on each drum: stroke time and strength	One tal cycle (16 beats of Teentaal)
Handloom (demo)	Whole-arm coordination and force in a steady rhythm	Beater motion, treadle switches, a phone photo of the cloth	One pick: open the shed, throw, beat

Craft	Kind of skill	What the tool sensor measures	The 'cycle' we compare
String puppetry (demo)	Fine, expressive finger control	A small motion sensor inside the puppet	One gesture phrase, e.g. a walk or a bow

A strike, a coordinated cycle and a fine expressive gesture are three different kinds of hand skill. If one sleeve and one pipeline handle all three, they can serve most hand crafts. The demos test only the two riskiest questions for each craft (E0 and E1), so the team's effort stays on the flagship.

What the jury will see live

1. **The sleeve.** The judge wears it, plays a tabla phrase or moves a Kathputli puppet, and feels which finger and which joint to move.
2. **The proof.** Real masters' fingerprints, and the test showing they belong to the master, not to the instrument or the day.
3. **Device off.** Cues fade, the judge performs alone, and the screen shows the unaided score.

Status today	What	Where
Working	PARAMPARA Lite: the teach, fade and device-off scoring loop on a phone, passing automated end-to-end tests (on time ≈ 100%, 60 ms late = 75%, silent = 0)	Section 5; PARAMPARA/lite
Working	Sleeve firmware core: drivers, address map, fade rule, scoring and clock sync, 27 unit tests passing on a PC	Section 5; PARAMPARA/firmware
Working	Analysis code run on real performers (10 drummers): performer information survives a new session and a new style; timing alone sits close to our pass mark	Section 7
Designed	Sleeve and tool kits: CAD files, datasheet-checked parts, budgets, eight defined bench tests (H1–H8)	Section 4; PARAMPARA/cad
Defined	E0–E5 with pass rules fixed in advance; partner kit and consent forms	Section 8; Appendices A–C
Not yet	A built sleeve, recordings of real tabla masters, E0–E5 results, signed partner letters. We claim none of these until they exist.	Weeks 1–8 (Section 11)

Our test of success: *the device succeeds only when the learner no longer needs it.*

2. The problem: three crafts, one gap

2.1 Skill that cannot be written down

All three crafts are learned by sitting with a master and imitating for years. Michael Polanyi called this tacit knowledge: 'we know more than we can tell' [19]; turning it into a shareable form is the hardest step in knowledge transfer [20]. Notation records the bols, a draft records the weave pattern, a script records the story. **None records the hands.**

2.2 The three crafts today

Craft	Situation	Source
Tabla	Taught face to face (<i>talim</i>). A school subject (CBSE 036) and a Kala Utsav category in 36 States/UTs. Ustad Zakir Hussain died in December 2024: his recordings stay, his teaching goes.	[33, 53, 54]
Handloom	35.22 lakh weavers and allied workers in 2019–20, down from 43.32 lakh. Training runs through 28 Weavers' Service Centres; 56,934 weavers trained under SAMARTH in six years. The national two-treadle qualification is 315 hours of hands-on practice. Handloom weavers report shoulder pain far more often than powerloom weavers (76% vs 42%).	[66, 67, 68, 69]
Puppetry	Four families of puppetry: string, shadow, rod and glove. In Kathputli the strings are looped on the puppeteer's fingers, with no control bar. Delhi's Kathputli Colony (about 2,800 families of folk artists) was moved to a transit camp in 2017.	[24, 71, 72, 73]
Policy	The State pays gurus to teach in person (Guru-Shishya Parampara), runs a scheme for intangible heritage, and counts about 64.66 lakh handloom and handicraft artisans. India has 16 elements on UNESCO's intangible heritage list.	[26, 27, 28, 30]

2.3 Why existing tools fall short

Tool	What it does well	What it cannot do
Video lessons and apps	Show the master's hands [34]	Measure the learner; check learning without help
Training centres and Guru-Shishya Parampara	The gold standard: a master in the room [28, 67]	Reach beyond one room, one master's time and one lifetime
Motion-capture and haptic suits	Measure body movement; US\$5,000 to over US\$12,000 [64, 65]	Teach a specific master's way at a school's budget; prove unaided learning
European craft-capture projects	Capture and represent craft gestures [74, 75]	Teach one master's fingerprint through a wearable that fades out

3. The innovation: a master's Skill Fingerprint, fingers to shoulder

3.1 What the fingerprint is

Layer	What it captures	Measured by
Body	How each finger, the wrist, elbow and shoulder move in each cycle: which joint leads, when, how fast and how far	9 motion sensors per arm
Tool	What the movement produces: drum stroke time and strength; beater timing and force, cloth density; the puppet's motion	One tool sensor per craft
Rhythm	Where in the cycle the master is slightly early or late, and how steady	Body and tool sensors together
Consistency	How much the master varies from cycle to cycle	Spread over at least 20 cycles

The fingerprint is the average and spread of these features over at least 20 cycles. **Only features that pass the confound test (Section 8) are used for teaching.**

3.2 Why the whole arm, not only the hand

Published finding	Meaning for us	Source
Expert pianists move shoulder, then elbow, then wrist in a clear sequence; novices do not	Expertise shows in how the joints are ordered	[56]
Experts let the arm's own motion do work at the elbow and wrist, so their muscles do less	Efficient technique is visible in the whole arm	[57]
Four drummers played the same accent with individual movement strategies	Movement carries personal style	[58]
Eight people were told apart from accelerometer gestures (98.6% recognition), used for user authentication	Movement alone can identify a person	[89]
Vibration at the joints in error cut real-time error by up to 27% and sped learning up to 23%	Joint-level touch cues work; that study needed an optical lab	[59]
76% of handloom weavers in a Varanasi survey reported shoulder pain	Measuring the shoulder also matters for health	[69]

3.3 Why we believe a fingerprint exists

Finding	Meaning for us	Source
Our pre-test on real performers: drummers named from timing and loudness on a new session and in a new style at about $3 \times$ chance; identical grooves, different drummers, 63% vs 25% chance	Performer information survives a change of day and of music; timing alone is close to our pass mark	Section 7 [88]
Tabla gharanas recognised automatically from audio of solo performances (38+ hours)	Style leaves a measurable trace	[39]
Systematic accent and tempo patterns within the tal cycle in a large Hindustani corpus	Accent and timing shapes are measurable	[38]
Pianists identified from timing and loudness; famous drummers each have a timing fingerprint	Performers can be told apart	[40, 41, 42]

3.4 What is new

Closest existing work	What it does	What it does not do
TIKL vibrotactile suit [59]	Joint-level vibration to correct arm poses	Use a low-cost wearable (it used an optical lab); capture fingers; teach a specific master; fade; test device-off
Teslasuit, Xsens suits [64, 65]	Body motion capture (and haptics)	Teach; prove the learner can perform alone; fit a school or weaving centre's budget
Mingei, CRAEFT [23, 74, 75]	Capture and represent craft gestures, simulators, haptics	A wearable that teaches one master's fingerprint and withdraws itself

Closest existing work	What it does	What it does not do
Pottery gesture capture, i-Treasures [21, 22]	Recognise expert gestures with body sensors	Teach them back with fading and device-off measurement
Haptic Drum Kit and bracelets [4, 5]	Vibration cues for which limb plays	Finger-level cues; a specific master's style; retention
Open-palm data glove [60]	Accurate finger tracking with the palm free	Teaching or a master reference

Our novelty is system-level. IMUs, haptics and performer recognition all exist. What does not exist, to our knowledge, is the integrated and validated workflow around one master's skill fingerprint:

Novelty statement

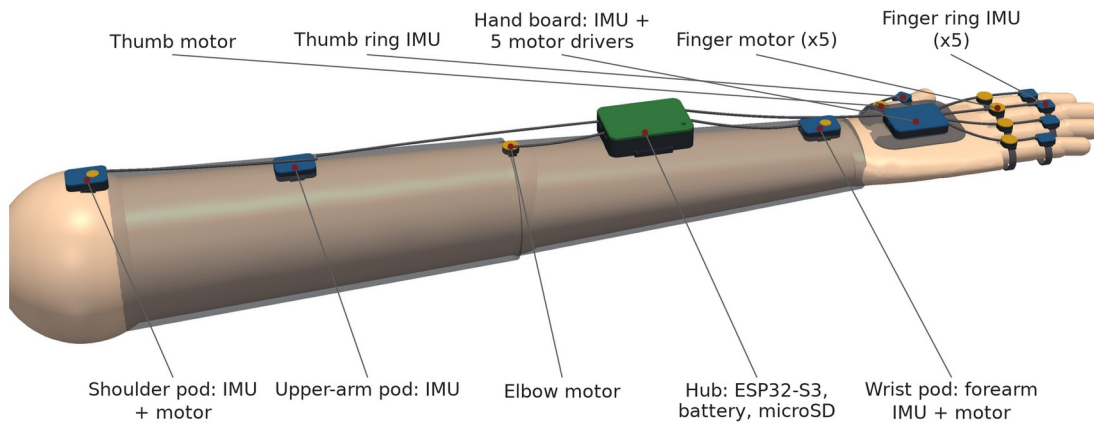
"To our knowledge, PARAMPARA is the first low-cost wearable that captures an individual master's skill fingerprint from fingers to shoulder together with what the tool does, proves that the fingerprint belongs to the master rather than to the instrument or the session, teaches it back through finger- and joint-level touch with sound and a ghost arm, withdraws the help as the learner improves, and counts only device-off performance, under the master's consent."

4. The equipment, exactly: drawings, datasheets, budgets and defined bench tests

Every view below is rendered from our CAD model (CadQuery, open source). Housing sizes are design values; electronic part sizes and ratings are taken from the manufacturers' datasheets. The arm, loom and puppet are stylised so the equipment can be seen; the equipment itself is drawn at its real size.

4.1 The sleeve at a glance

a Isometric view, right arm, palm down



b Dorsal (top) view, orthographic, to scale

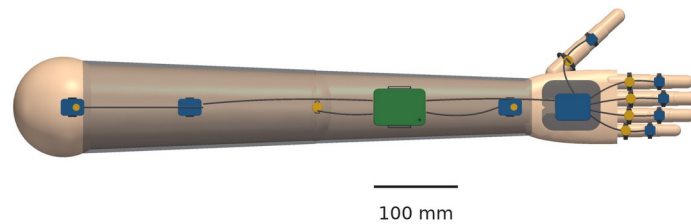


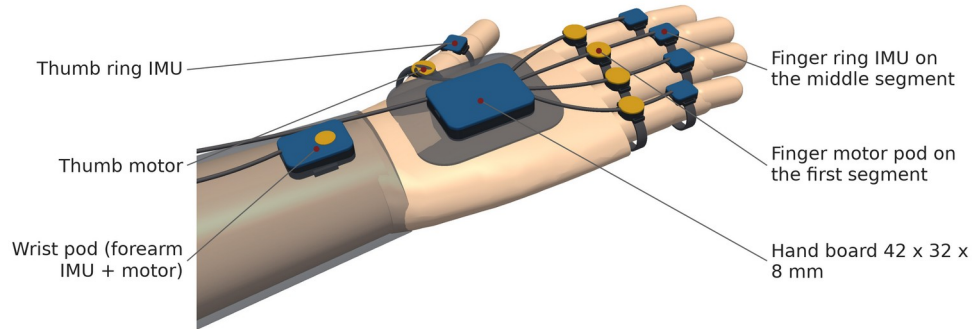
Figure 2. One sleeve per arm. (a) Isometric view. (b) Top view to scale. Blue lids are motion sensors, gold lids are vibration motors, green is the hub.

Unit (per arm)	Qty	Contents	Placement
Finger ring IMU pod	5	BMI270 on a 10 × 8 mm board	Middle segment of each finger, top of the thumb's first segment
Finger motor pod	5	8 mm coin LRA	First segment of each finger; thumb base
Hand board	1	BMI270, TCA9548A, 5 × DRV2605L	Back of the hand, on a fabric plate with a thumb loop
Wrist pod	1	BMI270 + LRA	Back of the lower forearm (measures forearm rotation)
Elbow motor pod	1	LRA	Back of the elbow
Upper-arm pod	1	BMI270	Middle of the upper arm
Shoulder pod	1	BMI270 + LRA	Top of the shoulder
Hub	1	ESP32-S3-MINI-1, TCA9548A, 3 × DRV2605L, charger, microSD, LiPo	Back of the upper forearm, in a sleeve pocket with a strap

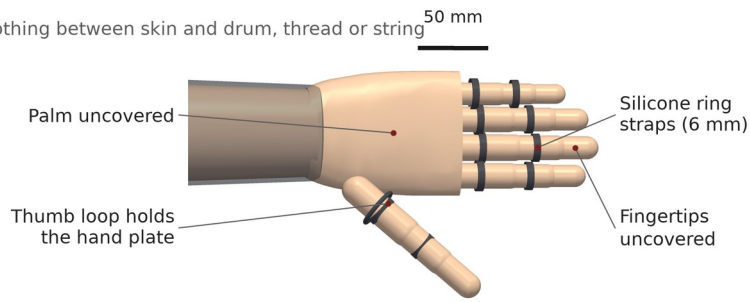
Per arm: 9 motion sensors and 8 vibration motors, wired to two I2C buses (Section 4.5).

4.2 The hand: fingertips and palm stay free

a Hand close-up



b Palm side: nothing between skin and drum, thread or string



c Side view (little-finger side), orthographic

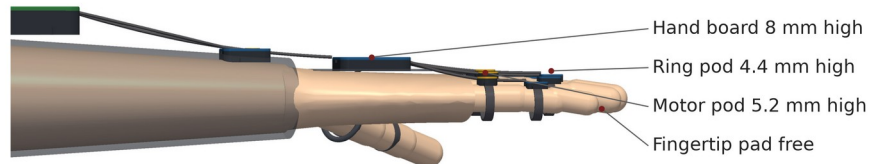


Figure 3. (a) Hand close-up. (b) Palm side: nothing covers the palm or fingertips, so the touch on the drum, the thread or the string is unchanged. (c) Side view: the tallest finger part is the 5.2 mm motor pod.

4.3 Inside each housing

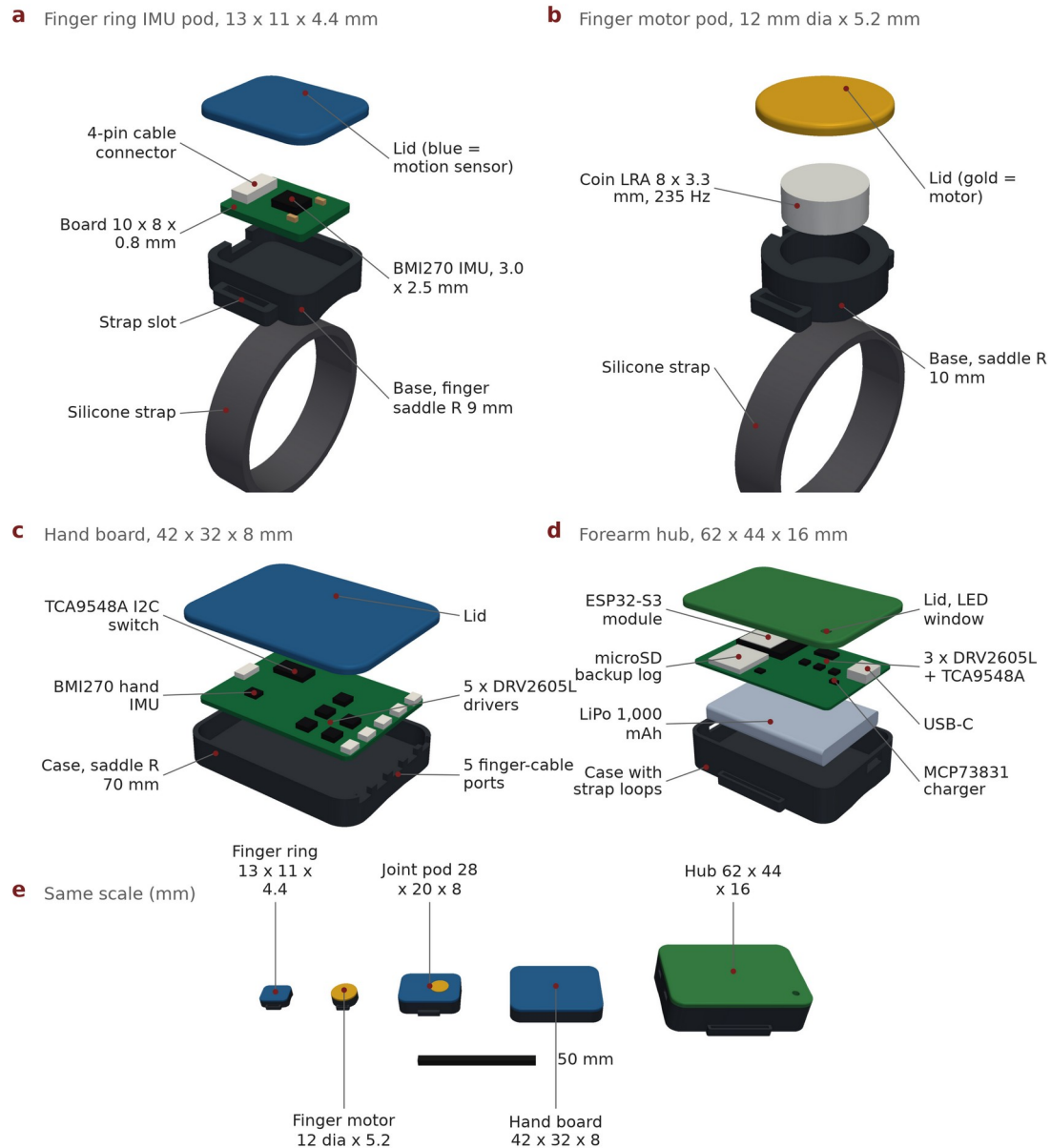


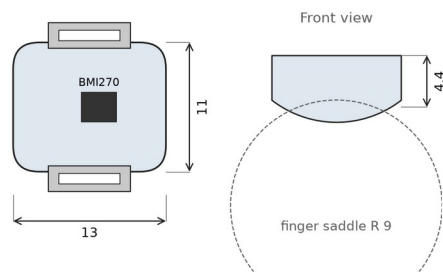
Figure 4. Exploded views. (a) Finger ring IMU pod. (b) Finger motor pod. (c) Hand board. (d) Forearm hub. (e) All housings to the same scale.

Part	Datasheet facts we rely on	Why this part	Source
BMI270 IMU	2.5 × 3.0 × 0.83 mm; 16-bit accelerometer and gyroscope; 685 µA; 2 KB FIFO; I2C addresses 0x68/0x69	Small enough for a finger ring; FIFO lets the hub read in bursts	[78]
DRV2605L haptic driver	2–5.2 V; 3 × 3 mm; closed-loop LRA auto-resonance; fixed I2C address 0x5A	Crisp, repeatable pulses whatever the motor's tolerance	[37, 79]
TCA9548A I2C switch	8 channels; 0x70–0x77; 100/400 kHz	Lets eight drivers with the same address share one bus	[80]
ESP32-S3-MINI-1	15.4 × 20.5 × 2.4 mm; dual core 240 MHz; Wi-Fi and Bluetooth 5	Two I2C buses, radio, enough speed for filtering	[81]

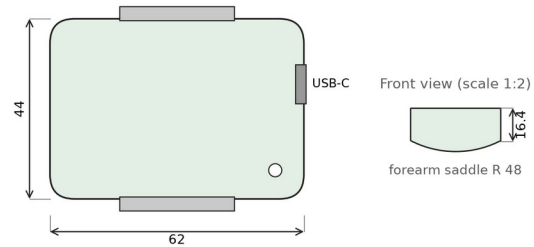
Part	Datasheet facts we rely on	Why this part	Source
C08-005 coin LRA	8 × 3.3 mm; 235 Hz; 1.8 V; 75 mA typical; 1.28 G	235 Hz sits where skin is most sensitive to vibration (Pacinian channel)	[16, 84]
MCP73831 charger	SOT-23-5; 15–500 mA programmable; 4.20 V	Safe single-cell charging over USB-C	[85]
7BB-27-4L0 piezo (tabla)	27 mm × 0.54 mm; 4.6 kHz; 20 nF	Contact pickup on the drum shell, no change to the head	[86]

4.4 Dimensioned drawings

Finger ring IMU pod: top view



Forearm hub: top view

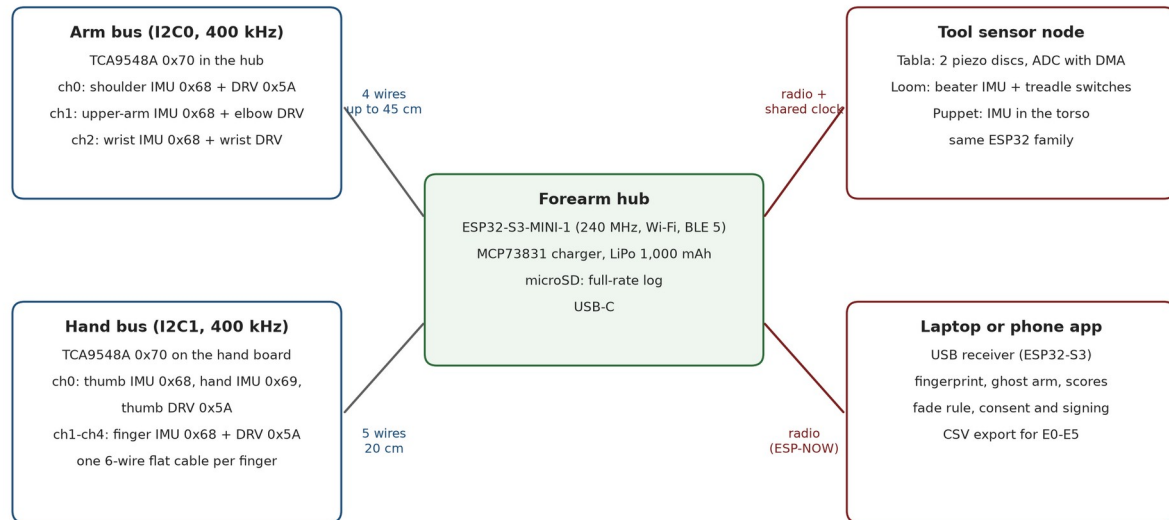


Layer (hub, bottom to top)	Height (mm)	Source
Case floor	2.0	design
LiPo pouch cell, 1,000 mAh, 50 x 34 x 6	6.0	cell size
Clearance	0.6	design
Main board	1.0	design
Tallest part: ESP32-S3-MINI-1 module	2.4	Espressif datasheet
Clearance to lid	2.0	design
Lid	2.4	design
Total	16.4	

Figure 5. Ring pod and hub with key dimensions (mm), and the hub's height stack-up.

4.5 Electronics

One sleeve: 9 IMUs (BMI270) and 8 vibration motors (coin LRA, each with a DRV2605L driver)



Each mux channel is its own short bus segment, so every segment stays far below the I2C limit of 400 pF.

Figure 6. Two I2C buses per sleeve. Each finger shares one switch channel between its ring IMU and its motor driver.

Why two buses and a switch on each. The DRV2605L has a fixed address, so eight drivers cannot share one bus without a switch [79, 80]. Each switch channel also isolates its own short cable segment: the longest (hub to shoulder, about 45 cm) adds well under 100 pF, far below the I2C limit of 400 pF per line [87].

4.6 Engineering budgets

Budget	Calculation	Result
Data (per sleeve)	9 IMUs $\times$ 12 bytes $\times$ 200 samples/s = 21.6 kB/s. Hand bus: 6 IMUs = 14.4 kB/s, about 36% of a 400 kHz bus; arm bus: 3 IMUs = 7.2 kB/s, about 18%	Fits, with FIFO burst reads [78]
Radio (two sleeves)	$2 \times 21.6 \text{ kB/s} \approx 350 \text{ kbit/s}$ to a USB receiver; a full-rate copy is also written to each hub's microSD card	Fits; no data lost if radio drops [Target]
Power (per sleeve)	ESP32-S3 with radio $\approx 110 \text{ mA}$ [82] + $9 \times 0.685 \text{ mA}$ IMUs [78] + motors 75 mA at $\leq 20\%$ duty $\approx 15 \text{ mA}$ [84] + microSD $\approx 10 \text{ mA} \rightarrow \approx 142 \text{ mA}$. $1,000 \text{ mAh} \times 80\% \text{ usable} \div 142 \text{ mA}$	$\approx 5.6 \text{ h}$ [Estimate]; pass mark $\geq 3 \text{ h}$ (H5)
Mass (per arm)	Housings from CAD volume at 1.01 g/cm^3 (PA12): ring 0.4 g, motor pod 0.4 g, joint pod 1.9 g, hand board 4.4 g, hub 15.4 g; plus boards, motors, 1,000 mAh cell ($\approx 20 \text{ g}$), cables and fabric	$\approx 160 \text{ g}$ per arm, $\approx 30 \text{ g}$ on the hand and fingers [Estimate]
Bus capacitance	45 cm of thin cable ($\leq 100 \text{ pF/m}$) + 2 devices ($\approx 10 \text{ pF}$ each) per channel	$< 100 \text{ pF}$ vs 400 pF limit [87]
Timing	Hubs and the tool node share a clock by timestamp exchange with drift correction [51]; tabla piezos sampled by DMA at 16 kHz per channel (driver allows up to 83 kHz) [83]	Body-to-drum sync $\leq 2 \text{ ms}$ [Target] (H3)

4.7 Tool kits

a Player's view

b Top view, to scale

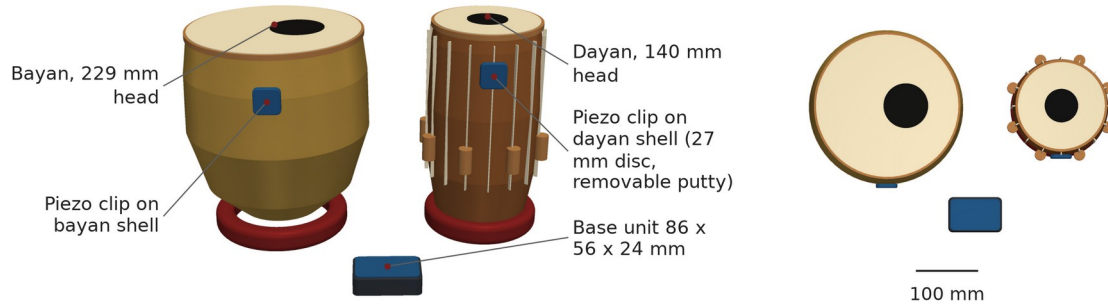


Figure 7. Tabla kit: a 27 mm piezo disc in a clip on each drum shell, held with removable putty, and a base unit. Drum sizes follow a standard 5.5-inch dayan and 9-inch bayan [90].

a Puppetry kit

b Handloom kit on a two-treadle frame loom

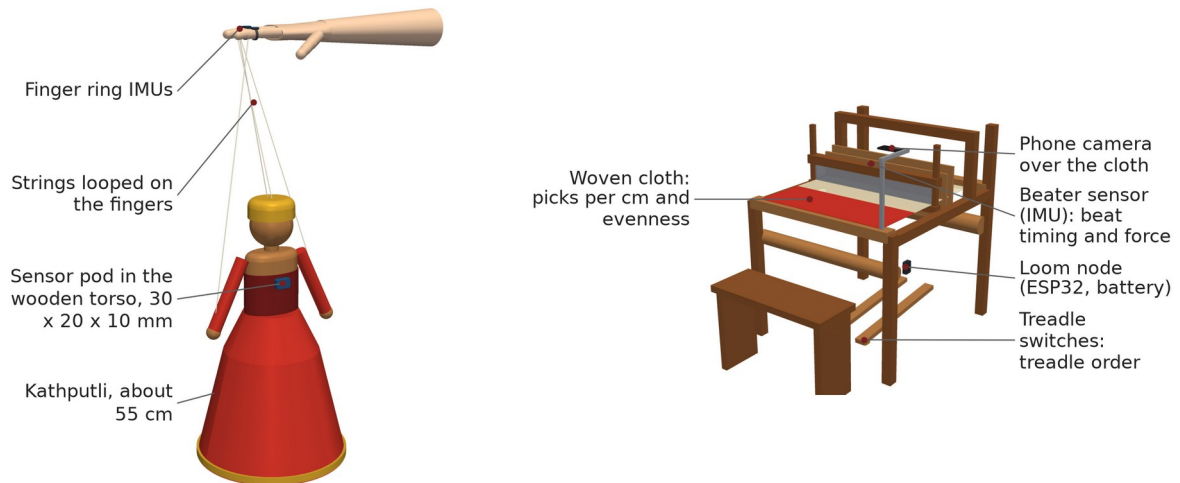


Figure 8. (a) Puppetry kit: a 30 × 20 × 10 mm sensor pod in the wooden torso of a Kathputli (typical puppets are 43–58 cm tall [91]); the strings stay on the fingers [71]. (b) Handloom kit on a stylised two-treadle frame loom.

Craft	Sensors	What they measure	Fitting
Tabla	Two 27 mm piezo discs [86]; base unit with ESP32-S3	Stroke time, strength and drum [18]	Clip on the shell, removable putty
Handloom	IMU on the beater; switches under both treadles; phone camera over the cloth; small loom node	Beat timing and force; treadle order; picks per cm and evenness [70]	Straps and clamps; nothing on the warp

Craft	Sensors	What they measure	Fitting
Puppetry	IMU pod in the torso; finger rings of the sleeve	Puppet's turn, tilt and movement; the puppeteer's fingers	Inside the costume; strings untouched

4.8 Defined bench tests, run before any person uses the sleeve

#	Test	Method	Pass [Target]
H1	Sampling	Logic analyser on each bus; 10 minutes at 200 samples/s	Interval jitter ≤ 0.5 ms; no missed samples
H2	Cue latency	Accelerometer taped to a motor pod; command to vibration onset, 500 cues	Latency ≤ 20 ms; jitter ≤ 3 ms
H3	Body-to-drum sync	Tap the drum head with a sleeved finger 200 times; compare IMU spike with piezo onset	Difference ≤ 2 ms in 95% of taps
H4	Data integrity	Two sleeves streaming for 30 minutes; compare radio log with microSD log	Radio loss $< 1\%$; microSD complete
H5	Battery	Continuous streaming with cues at 20% duty	≥ 3 hours
H6	Orientation	Pods on a 0/30/60/90° jig, static	Error $\leq 2^\circ$
H7	Fit and comfort	Three hand sizes; donning time; pod temperature after 1 hour; weigh the sleeve	Donning ≤ 3 min; warming $\leq 5^\circ\text{C}$; ≤ 200 g per arm
H8	Stroke detection	200 tabla strokes labelled from slow-motion video	Precision and recall $\geq 95\%$

4.9 Safety, hygiene and materials

- **Low voltage only:** one 3.7 V cell per sleeve with a protected pouch cell and a charger with thermal regulation [85]; no mains connection while worn.
- **Skin contact:** medical-grade silicone straps; PA12 housings with rounded edges; fabric sleeve washable once pods clip out.
- **Comfortable vibration:** cue strength set from each person's own threshold (E2), with a hard maximum in firmware.
- **Weight on the hand about 30 g [Estimate]:** E0 checks that it does not change how masters play.

4.10 Open design files

STEP and STL files for every housing, the CAD script and the render script are in PARAMPARA/cad (Appendix G). Anyone can print, inspect or improve them.

5. Working today: phone prototype, firmware core and the proof ladder

Three parts of PARAMPARA run today without the sleeve. They do not replace the sleeve, the masters or E0–E5; they remove risk from them and let us start collecting data this week.

5.1 PARAMPARA Lite: the teaching loop on a phone

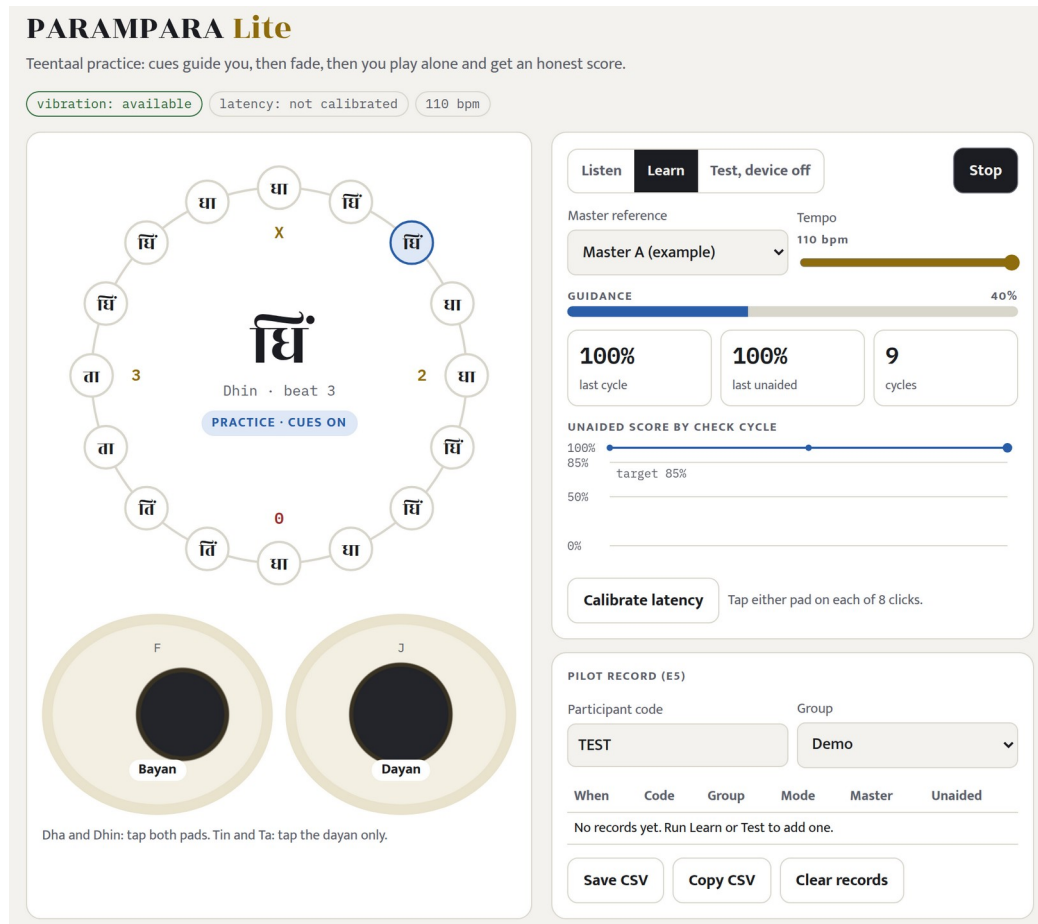


Figure 9. PARAMPARA Lite during an automated test session: the tal dial with sam (X), tali (2, 3) and khali (0), the two pads, guidance faded to 40% after three check cycles, and the unaided score per check cycle. Screenshot of running software; Master A is an example profile.

- **What it does:** plays the master's Teentaal with the master's own early and late timing; gives cues just before each stroke (phone vibration on Android Chrome, a glowing pad, and the master's sound, which gets quieter as guidance falls); runs check cycles with no help; fades guidance with the same rule as the sleeve; scores the device-off test; stores pilot records and exports a CSV.
- **Latency calibration:** eight clicks, and the median tap delay is removed from every score, so phone delay does not count as a timing error.
- **Where:** a private web link for the team to share, and `PARAMPARA/lite/index.html`, which opens directly on a phone.

Automated end-to-end test (headless Chromium, scripted player)	Unaided score	Guidance after 2 checks	Result
Player taps exactly on the master's timing	99.96%, 99.98%	100% → 60%	PASS
Player 60 ms late on every stroke (rule gives 75%)	75.0%, 74.9%	100% → 60%	PASS
Silent player	0%	stays 100%	PASS

The test drives the page's own tap path while the real audio clock, scheduler, scoring and fade rule run, so it checks the software, not a person. Code: `PARAMPARA/lite/e2e_test.py`.

E5a: a learning pilot we can start this week

Sixteen beginners are randomised to **check-cycle fading** or **always-on cues** on PARAMPARA Lite, practise in three short sessions, and take the device-off test 48 hours later. This tests the fade rule on people instead of simulations (Appendix E) before the sleeve exists. It is a pilot: it estimates the effect with a 95% confidence interval and does not prove better learning.

5.2 Sleeve firmware core, tested on a PC

Unit tests (27 checks, all passing)	What they prove
Address map	9 IMUs and 8 motor drivers: every (bus, switch channel, address) is unique
Bring-up on mock buses	All 17 devices initialised; no device is touched without a switch channel selected
DRV2605L set-up	LRA mode, LRA library, real-time playback; rated-voltage code 70 for 1.8 V rms at 235 Hz [79, 84]
BMI270	200 Hz configuration after the configuration upload; init refuses a wrong chip ID [78]
Motor commands	Right channel, and no redundant switch writes
Fade, cues and scoring	Guidance falls by at most 0.2 per check and reaches 0 with sustained good play; cue density follows guidance; a 60 ms late cycle scores 0.75
Clock sync	50 ppm drift, 2 ms offset, 200 μ s jitter and 5% outliers: worst error 63 μ s over two minutes (target ≤ 2 ms, H3)

The same headers compile into the ESP32-S3 build in week 1; the target toolchain could not be downloaded in our build environment, so the target build itself is not yet verified. Bosch's BMI270 configuration file comes from its open-source sensor API.

5.3 The proof ladder

#	Rung	Status	What the jury can check
1	Idea, design and validation plan with pass rules fixed in advance	Done	This document
2	Method pre-tested on real performers	Done	Section 7, code and data source
3	Teaching loop working on a phone, with end-to-end tests	Done	PARAMPARA Lite link; test log
4	Sleeve firmware core with unit tests	Done	27 passing tests
5	Sleeve built; H1–H8 bench tests pass	Weeks 1–3	Bench-test sheet with measurements
6	Tabla masters recorded with consent; E0 shows the sleeve does not change their playing	Weeks 4–5	Signed consent; E0 equivalence result
7	E1 verdict: the fingerprint belongs to the master	Week 5	T-a, T-b, T-c and the claim it allows (Section 8.1)
8	E5a and E5: device-off learning estimates	Weeks 1–7	Confidence intervals, published either way

Every rung we reach is shown at the finale with its measurement. Every rung we miss is shown as missed.

6. How it works: Fingerprint, Teach, Fade, Measure, Own

6.1 Fingerprint

The tool sensor marks each cycle: a tal cycle on tabla, a pick on the loom, a gesture phrase on the puppet. For every cycle the software removes the cycle's overall tempo and size, then stores, for each joint, **when it moves relative to the tool event, which joint leads, how fast and how far**, with the tool features and the rhythm. Averaging over cycles gives the fingerprint; the spread shows the master's consistency. Orientation comes from the Madgwick filter [61] after a 20-second N-pose calibration [63]. Because shoulder angles from body-worn sensors can be off by 10–20° [62], the fingerprint relies mainly on timing, order and speed, and E1 re-fits the sleeve on each day so placement error is tested, not assumed away.

6.2 Teach: each sense does what it is best at

Sense	What it carries	Why
Touch: finger and joint motors	Which finger moves next; which joint should lead or is lagging; how strong the accent is	Joint-level vibration improved arm learning [59]; drum and strength cues were recognised 96% of the time [7]
Hearing	Exact timing: the master's tiny early and late shifts	The ear notices about 10 ms [43]; the skin needs 10–30 ms or more [46]
Sight: ghost arm	The master's arm as a moving figure over the learner's; replay after each cycle	Visual feedback suits movement shape and review [10]

Timing the cue. Touch is dulled on a limb that is about to move [44, 45], so each cue ends just before the movement and is set well above the learner's threshold.

Level	What the learner learns	Moves on when
1. Structure	The pattern: which finger, which joint, which stroke or step	Performs it without cues, 80% correct
2. Steady rhythm	Keeping the cycle steady while cues fade out	Unaided score $\geq 85\%$ on two different days
3. Style	This master's own timing, accents and joint order	Matches the master's fingerprint and the master approves (only where E1 passes)

6.3 Fade

Feedback given less often leads to better long-term learning, even though practice feels harder [13, 14]. At the start of each session and every four cycles the learner performs one cycle with **no cues**. Better check cycles mean fewer cues; two struggles in a row bring cues back a little; reaching the target on two different days asks the master to approve the next level.

6.4 Measure: with the device off

Learning means what the learner can do **after a break and without help** [12].

Craft	Unaided score (device silent)
Tabla	Timing, accent and stroke accuracy against the master's reference; steadiness and early/late tendency reported separately [15]
Handloom	Evenness of picks per cm (phone photo), steadiness of the beat, correct treadle–throw–beat order; time with the shoulder raised reported for health
Puppetry	How closely the puppet's motion follows the master's (after time alignment), and whether viewers pick it as the master's

6.5 Own: consent and credit stay with the master

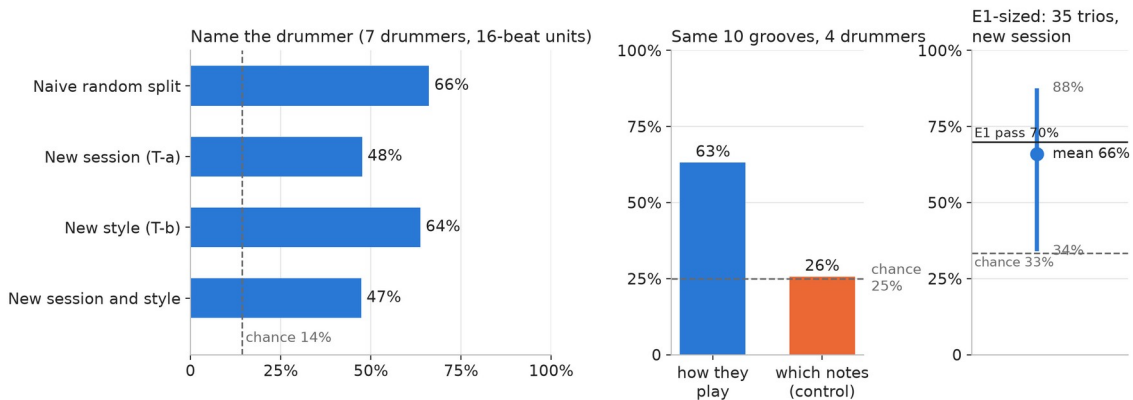
- The master decides how the recording may be used (teaching, sharing, commercial use, time limit) and can withdraw it.
- The master approves each recording and each learner's progress with the fingerprint sensor on their own phone; the key never leaves the phone [49].
- Credit and terms travel with the data as machine-readable labels [48]; an edited copy fails verification.
- This follows UNESCO's ethical principles (free, prior and informed consent) and the CARE principles [25, 47].

7. Pre-test on real performers: ten drummers

This is **not** a test of the invention. It checks one assumption behind it (that a performer's playing carries information that survives a new day and a new style) on real performers, before any master is recorded. We could not record tabla masters before this document, and no public multi-player tabla dataset was reachable, so we ran our confound-controlled method on the closest public data: the **Groove MIDI Dataset**, 1,150 performances by 10 drummers recorded to a click on one electronic kit over several sessions, with drummer, session and style labels (CC BY 4.0) [88]. Four drummers also played the **same ten written grooves**: the drum version of 'every master plays the same theka differently'.

7.1 Method, fixed before running

- **Unit:** 16 beats (four bars), the length of one Teentaal cycle; 4,118 units from the 7 drummers with at least two sessions. One-bar units were also run and are reported.
- **Features (how, not what):** for kick, snare and hi-hat/ride, the timing against the click and the loudness on beats, offbeats and sixteenths, plus their spread: 24 numbers per unit.
- **Classifier:** the same simple nearest-centroid method as our E1 code; balanced accuracy, so chance is one over the number of drummers.
- **Tests:** new session; new style (a style never seen in training); both at once; same grooves; a control using only which notes were written; and the direct confound test with 1,000 label shuffles.



Real data: Groove MIDI Dataset (Gillick et al. 2019, CC BY 4.0), 4,118 sixteen-beat units, 7 drummers. Features: timing against the click and loudness only.

Figure 10. Real data. Left: drummers named from 'how they play' on a new session and in a new style at about three times chance. Middle: identical grooves, four drummers: 'how' names the drummer (63%), 'which notes' does not (26%, chance 25%). Right: in E1-sized groups of three drummers, the new-session test averaged 66%.

Test (real data)	16-beat units	1-bar units	Chance
Naive random split (inflated)	66%	59%	14%
New session (T-a)	48%	43%	14%
New style (T-b)	64%	56%	14%
New session and new style together	47%	44%	14%
Same ten grooves, 4 drummers: how they play	63%	63%	25%
Same ten grooves: which notes only (control)	26%	26%	25%
Same grooves, whole performances named correctly	26 of 40	27 of 40	10 of 40
Direct confound test: gap (p, 1,000 shuffles)	+0.73 (p = 0.001)	+0.54 (p = 0.001)	gap ≤ 0

Test (real data)	16-beat units	1-bar units	Chance
E1-sized trios, new session: mean (range)	66% (34–88%)	63% (34–83%)	33%
E1-sized trios reaching our 70% pass mark	13 of 35	11 of 35	

7.2 What this proves, and what it does not

What it proves

- **Performer-specific information in timing and loudness** survives a new recording session and a different musical style, at about three times chance.
- **It lives in how people play, not what they play:** with identical grooves, notes alone are at chance while timing and loudness name the drummer.
- **Our method and code work on real performers**, not only on synthetic data.

What it does not prove, stated plainly

- It does not demonstrate PARAMPARA: it is not tabla, not Indian masters and not body movement, but MIDI from Western drummers on one electronic kit, so it cannot test an instrument swap.
- **Timing and loudness alone sit near our pass mark:** in groups of three, the new-session test averaged 66%, and 13 of 35 trios reached 70%. A tabla E1 based only on sound could fail.
- That is exactly why PARAMPARA adds the sleeve: movement carries individual strategy [56, 58, 89]. **We keep the 70% pass mark unchanged.**

```
python3 PARAMPARA/sim/gmd_fingerprint.py groove/ # reproduces Section 7
```

8. Validation plan: E0–E5

8.1 The hardest question

"Are you identifying the master, or their instrument, tuning, recording session or sensor placement?"

Music systems that appeared to recognise artists were partly recognising each album's recording conditions, the 'album effect' [76]. So E1 breaks that link. **Per craft:** 3 masters × 2 sessions on different days (sleeve re-fitted) × 2 instruments (masters swap tablas, looms or puppets) × 20 cycles = **240 cycles**.

Test	Question	Pass [decided in advance]	Rules out
T-a New day	Trained on one day, does it name the master on the other day?	≥ 70% (chance 33%)	Session, sensor placement
T-b New instrument	Trained on one instrument, does it name the master on the other?	≥ 70% (chance 33%)	Instrument, tuning
T-c Direct test	Same master on different instruments more alike than different masters on the same instrument?	Yes, $p < 0.01$ (1,000 shuffles)	All of the above

Test	Question	Pass [decided in advance]	Rules out
Control	How well do the same features name the instrument?	Reported	How much instrument signal remains

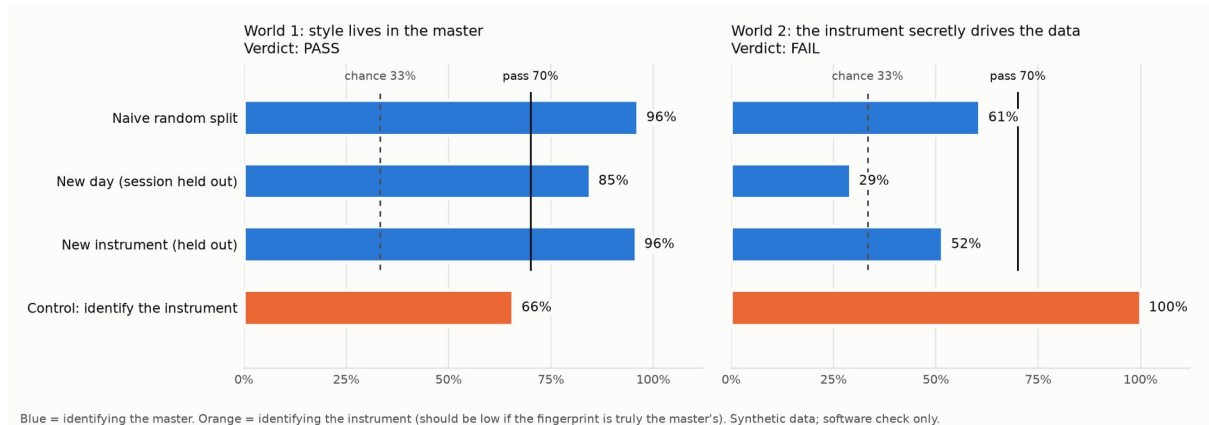


Figure 11. Software check on synthetic data: when the instrument secretly drives the data, a naive split still looks above chance (61%), but the new-day and new-instrument tests fall to 29% and 52% and the verdict is FAIL.

What each E1 result lets us claim (fixed in advance)

E1 result for a craft	What we claim	What we teach
T-a, T-b $\geq 70\%$ and T-c $p < 0.01$	A master-specific style fingerprint	Levels 1–3
T-c $p < 0.01$, but T-a or T-b between 50% and 70%	A real but weak fingerprint; not enough to teach one master's style	Levels 1–2; style shown for information only
T-c not significant	No master fingerprint for this craft	Levels 1–2

8.2 All experiments

#	Question	Design	People	Pass [Target]	Crafts
E0	Does the sleeve change how masters perform?	Alternating blocks of 10 cycles with and without the sleeve; the tool sensor measures both	Each master	Equivalent within ± 10 ms, $\pm 5\%$ cycle time, ± 1 dB (two one-sided tests [77]); comfort ≥ 4 of 5	All three
E1	Is the fingerprint the master's?	Section 8.1	3 master s per craft	T-a, T-b $\geq 70\%$; T-c $p < 0.01$	All three
E2	Can cues be felt while performing?	Weakest felt cue at rest vs while performing; finger, wrist, elbow, shoulder	8	At most twice the rest strength at one site per joint	Tabla, puppetry
E3	Can people tell two masters apart through PARAMPARA?	A, B, then X; 20 rounds	10	≥ 15 of 20 ($p = 0.02$)	Tabla
E4	Can cues be read while performing?	40 random cues: 5 fingers, 3 joints, 2 strengths	8–10	$\geq 90\%$ sites, $\geq 80\%$ strengths	Tabla, puppetry

#	Question	Design	People	Pass [Target]	Crafts
E5a	Does fading beat always-on cues, on a phone?	PARAMPARA Lite: 16 people randomised to check-cycle fading or always-on cues; 3 sessions; device-off test 48 h later	16	Difference with 95% CI (a pilot estimate)	Tabla, this week
E5	Do learners improve with the device off?	16 beginners randomised: video practice or PARAMPARA with the sleeve; 3 sessions; retention test 48 h later, device off, blinded tester	16	Difference with 95% CI; full trial if PARAMPARA $\geq$ video	Tabla

Honest sample sizes. With 8 people per group, E5a and E5 can only detect a very large effect ($d \approx 1.5$ at 80% power). They are pilots that estimate whether PARAMPARA improves unaided retention; **we do not claim that it improves learning until the full trial.** The 120-person trial in the roadmap (60 per group) detects $d \approx 0.52$ [92]. All studies have written consent, institute ethics approval, and pass rules registered before recording.

9. What we build for SIH

9.1 Flagship and demos

Craft	Validated at SIH	Live at the finale
Tabla (flagship, $\approx$ 80% of effort)	H1–H8, E0, E1, E2, E3, E4, E5a, E5	Judge plays with the sleeve; two masters' fingerprints with the E1 verdict; device-off score
Puppetry (demo, $\approx$ 10%)	E0, E1, E2, E4	Judge moves a Kathputli and feels the string-finger cues
Handloom (demo, $\approx$ 5%)	E0, E1	A weaver's recording replayed on the ghost arm; cloth evenness from a phone photo

9.2 Three deliverables

Deliverable	What the judge experiences	Done when [Target]
1. Two sleeves and three tool kits	Wears a sleeve and feels finger and joint cues while performing	H1–H8 pass; $E4 \geq 90\%$
2. Proven fingerprints	Real masters' fingerprints with the confound-test verdict for each craft	E0 and E1 reported, pass or fail
3. Device-off learning	Practises; cues fade; performs alone; sees the unaided score and the E5 result	Fade runs end to end; E5 reported with its CI

9.3 Bill of materials (two sleeves, three tool kits)

Item	Qty	Approx. ₹ [Estimate]	Source
BMI270 IMUs (US\$1.26 each at volume)	18	2,000–3,000	[52]
DRV2605L drivers (US\$0.72 each at volume)	16	1,000–2,000	[52]
TCA9548A switches	4	300–600	[80]
ESP32-S3-MINI-1 modules (hubs) and ESP32 boards (tool nodes, receiver)	2 + 4	2,500–3,500	[81]
Coin LRAs 8 mm (US\$5.15 at 1–9 units; cheaper in volume)	16	3,000–7,000	[52, 84]
LiPo 1,000 mAh cells, MCP73831 chargers, microSD cards	2 sets	1,500–2,200	[85]
Circuit boards made and assembled by a PCB service (ring, joint, hand and hub boards)	≈ 20 boards	4,000–6,000	–
3D-printed PA12 housings, silicone straps, fabric sleeves, flat cables	2 sleeves	3,000–4,000	–
Tool kits: piezo discs and base unit; beater IMU and treadle switches; puppet pod	3	2,500–3,700	[86]
Total prototype		≈ 20,000–32,000	

US\$ prices converted at about ₹85 per US\$. A whole kit costs less than one Teslasuit (about US\$5,000) by a factor of more than ten [64]; at volume, one sleeve could cost about ₹5,000–7,000 [Target].

10. Evidence from published research

Finding	Key number	Source
Joint-level vibration suit while copying a teacher	Error down up to 27%; learning up to 23% faster	[59]
Haptic guidance added to audio training (drumming task)	–17% loudness error; –18% early timing error	[1]
Haptic vs visual training of a movement	Timing learned better from haptics	[3]
Beginners learned drum patterns from vibration alone	Haptic Drum Kit	[4]
Violin bowing with vibration feedback	Improved; half kept the gain without feedback	[6]
Feeling a drum and strength cue on the body	96.18% recognised	[7]
Piano: vibration vs visual cues (n = 14)	Timing error 12.1% vs 22.3%	[8]
Feedback on half the trials vs every trial	Better retention	[14]
Low-cost orientation filter	Below 0.8° static, 1.7° dynamic	[61]
Open-palm glove, fingertips free	Finger error 1.25°	[60]

What we will not claim

- That PARAMPARA transmits a master's 'soul'. We measure and teach **observable** parts of their skill.
- That it replaces the master. The master records, approves and controls.
- That a 16-person pilot proves long-term learning, or that the sleeve reduces weavers' pain.
- That the drummer test proves a tabla fingerprint. It shows the method works on real performers and calibrates our risk.

11. Feasibility

11.1 Eight-week plan

Week	Build and record	Proof produced
1	Order boards from a PCB assembly service; print housings; first sleeve on breakout boards. Partner outreach (Appendix A); ethics application; pass rules registered. Start the E5a phone pilot with PARAMPARA Lite.	H1 on breakout boards; E5a sessions
2	Hub firmware: two buses, FIFO reads, motors, microSD, radio; tabla kit	H2, H3, H8
3	Assembled boards arrive (fallback: breakout boards in larger rings); both sleeves built	H4–H7; E2 and E4 on volunteers
4	Fingerprint pipeline and ghost arm; tabla masters, day 1	E0 (tabla); E1 day 1
5	Tabla masters, day 2 (instruments swapped, sleeve re-fitted); puppeteers, days 1–2	E1 verdict (tabla); E0 and E1 (puppetry); E3
6	Weavers at a weaving centre, days 1–2; learning pilot starts	E0 and E1 (handloom)
7	Learning pilot: 16 beginners, 3 sessions each, retention test	E5 with confidence interval
8	Full demo; three rehearsals without help; backup video; deck with real numbers	Demo runs end to end

11.2 Team of six

Role	Owns
Hardware	Boards, housings, sleeves, tool kits, bench tests H1–H8
Firmware	Buses, cue timing, radio, clock sync, logging
App	Ghost arm, lessons, scores, consent screens
Analysis	Fingerprint features, confound test, fade rule, statistics
Research and partner liaison	Outreach to masters, consent, recording sessions
Pitch	Deck, demo script, backup video

11.3 Top risks

Risk	What we do	Fallback
No masters available in time	Outreach in week 1 through Zonal Cultural Centres, Weavers' Service Centres, music colleges and puppeteer communities	Senior practitioners of different teachers, stated clearly
Assembled boards late	Order in week 1; breakout boards from day 1	Larger rings on breakout boards for E0–E4
Sleeve changes how masters perform (E0)	Lighter rings; adjust positions with the master	Hand, wrist and elbow sensors only
Fingerprint near the pass mark (as in Section 7)	Add movement features; record 30 cycles	Graded claim (Section 8.1)
Finger cues not felt while striking (E2)	Cue before the movement; stronger pulse	Move the cue to the wrist
Radio trouble at the venue	microSD log; cues stored on the sleeve	USB cable; backup video

12. Viability: institution-first, craft by craft

Our first customers are institutions, because hardware cost and the need to protect masters' content make direct consumer sales premature.

Craft	What already exists	What PARAMPARA adds
Tabla	Guru-Shishya Parampara honoraria (e.g. ₹7,500 a month for a guru) [28]; CBSE subject 036 [53]; ABGMVM exam centres [31]	A kit per guru or music room; fingerprints as lasting output of state-funded teaching
Handloom	28 Weavers' Service Centres and SAMARTH training [67]; a 315-hour NSQF qualification [68]	A training aid with device-off scores as evidence of skill; shoulder feedback for weaver health
Puppetry	Intangible heritage scheme through Sangeet Natak Akademi [27]; artists in schools [55]; artists-in-residence [29]	Recorded, credited fingerprints of puppeteers; a teaching kit for workshops
All	National audiovisual archive [50]	A new kind of record: how masters move

- **Cost:** about ₹20,000–32,000 for the two-sleeve prototype; less than four months of a guru's state honorarium [28] and a small fraction of a motion-capture suit [65].
- **Who pays:** institutions buy kits and a yearly licence; consenting masters receive a share whenever their fingerprints are used; grants and CSR support preservation. **These are hypotheses: our first-year test is three signed letters of intent.**
- **Rules:** explicit, recorded consent; data stays on the device or the user's app by default [35]; performers' rights respected [36].

13. Impact

Who	Benefit	How we will measure it
Masters	Their own way of playing, weaving or puppeteering preserved, credited and shared on their terms	Masters recorded; their feedback
Learners	Practise against a real master between lessons, and see honest progress	Unaided score after a break
Weavers	Feedback on shoulder posture during long hours [69]	Time with the shoulder raised (to be tested)
Archives	A record of how masters move, not only how they sound and look	Consented fingerprints deposited
Researchers	A consented dataset linking masters' fingerprints with learners' progress	Dataset and publications

Possible wider benefit, not yet tested: deaf people can follow a beat through vibration [9]; India has about 50.7 lakh people with hearing disability [32].

14. The pitch

14.1 Six things the jury should remember

1. A master's skill lives in the fingers, wrist, elbow and shoulder.
2. We measure it with one low-cost sleeve; tabla is proven, handloom and puppetry show it generalises.
3. We test whether the fingerprint is the master's and not the instrument's; on real drummers our method already finds performer information.
4. We teach it through touch, sound and a ghost arm.
5. We remove the help and count only what the learner does alone.
6. The master owns and controls the recorded legacy.

14.2 Three-minute demo

Time	What happens
0:00–0:20	'Archives keep how masters sound and look. Nothing keeps how their hands move.'
0:20–0:50	Two tabla masters' fingerprints and the E1 verdict: still recognised on a new day and on a swapped tabla.
0:50–1:50	The judge wears the sleeve and plays along. Cues fade. A check cycle runs with the device silent; the unaided score appears. 'The device succeeds only when it is no longer needed.'
1:50–2:20	The judge moves a Kathputli and feels the string-finger cues; a weaver's recording replays on the ghost arm.
2:20–2:40	The master's consent: approved with a phone fingerprint; an edited copy is rejected.
2:40–3:00	E0–E5 results and the drummer test; institutions first.

14.3 Hard questions

Question	Answer
Are you identifying the master, or the instrument, tuning, session or sensor placement?	We test exactly that: two days with the sleeve re-fitted and two instruments. A fingerprint counts only if it names the master on a new day and a new instrument (Section 8). On real drummers, the method already separates performer from session and style (Section 7).
Show me proof, not plans.	What works today: the phone prototype (try it), the firmware core (27 tests), and the analysis on real performers. What is designed: the sleeve, with eight defined bench tests. What is not yet done: the sleeve build, master recordings and E0–E5, each with pass rules fixed in advance and reported whichever way they go.
Nice CAD. Where is the sleeve?	It is built in weeks 1–3 on breakout boards, then on assembled boards. Its first test is E0: does it change how the master plays? Until then, the teaching loop runs on a phone and the firmware core is tested on a PC.
Have you proven the fingerprint?	Not for tabla yet. On real drummers, performer information survives a new session and style, but timing alone is close to our pass mark. That is exactly why we add body motion, and why E1 is a real test.
Three crafts: isn't that spreading thin?	Tabla is the flagship and gets every test. Handloom and puppetry get only E0 and E1, to show the platform generalises.
Doesn't the sleeve change how a master plays?	Fingertips and palm stay free, about 30 g sits on the hand, and E0 measures performance with and without it.
Does PARAMPARA teach better than video?	Not yet shown. E5 is a randomised pilot with a device-off retention test; the 120-person trial follows.
Why not a motion-capture suit?	Suits cost US\$5,000 or more and do not teach [64, 65].
Can vibration carry tiny timing?	No, and we don't ask it to: touch carries which finger, which joint and how strong; the ear carries timing [43, 46].
Can we meet the masters?	They are named in the deck with signed consent; with their agreement we invite them or share their contact through our institute.
What if E1 fails?	We follow the graded claim table in Section 8.1 and say so.

15. Limits we state openly

- **Body sensors have limits:** shoulder angles can be off by many degrees [62]; we rely on timing, order and speed.
- **The real-performer pre-test is not tabla:** Western drummers, MIDI, one kit; it cannot test an instrument swap and does not demonstrate PARAMPARA.
- **PARAMPARA Lite is a phone stand-in:** example master profiles, not real masters; taps carry no stroke strength; phone vibration may be blocked inside some viewers.
- **Timing alone is near the pass mark** (Section 7); the style claim depends on the sleeve's movement data.
- **Not captured:** grip force inside the hand, tone colour, thread tension.
- **Small first studies:** E0–E5 test feasibility and estimate effects.
- **Looms are not portable:** handloom work happens at weaving centres; the finale shows recordings.

- **Some masters may not want to be recorded.** Their choice decides what we do.

16. Roadmap after SIH

When	What	Starts only if
0–6 months	Tabla learning trial with about 120 people (detects $d \approx 0.52$) [92]	E0–E5 pass for tabla
0–6 months	Handloom training pilot at a Weavers' Service Centre, with shoulder feedback	E0 and E1 pass for handloom
0–6 months	Flexible-circuit sleeve; volume cost about ₹5,000–7,000; on-device stroke recognition [17]	Bench tests stable
6–12 months	Style level where E1 passes; archive pilot with NCAA [50]	Clear fingerprints in E1
12+ months	Glove and shadow puppetry, kathak footwork, pottery [22, 72]	Three-craft deployment working

17. References

Status at the end of each entry: **V** verified by live web search in October 2026; **P** existence verified, some details from memory (check before quoting); **S** standard textbook or classic paper. Only references cited in this document are listed.

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I. Hardware, platforms and market

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J. Style, micro-timing and performer identification

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K. Touch during movement and tactile timing

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L. Cultural data governance, identity and archives

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M. Engineering, cost and access

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N. Prior art and education channels

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- [55] NEP 2020 para 4.26 and the PSSCIVE guidelines for 10 bagless days (2024): students in Grades 6–8 intern with local vocational experts such as carpenters, gardeners, potters and artists. **Status: V.** https://psscive.ac.in/storage/uploads/documents/1722421697_Implementation%20of%2010%20Bagless%20Days%20in%20School.pdf

O. Whole-arm skill and wearable motion sensing

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- [57] Furuya, S., & Kinoshita, H. (2008). Expertise-dependent modulation of muscular and non-muscular torques in multi-joint arm movements during piano keystroke. *Neuroscience*. Experts produced larger interaction torques at the elbow and wrist, so their muscles did less work than novices'. **Status: V.**
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- [60] Hosie, O., Isaksson, M., McCormick, J., Tirosch, O., & Hensman, C. (2025). The design and validation of an open-palm data glove for precision finger and wrist tracking. *Sensors*, 25(2), 367. DOI 10.3390/s25020367. Fingertips and palm left free; static mean absolute error 1.25° for the finger sensors and 4.85° for the wrist IMU. **Status: V.** <https://www.mdpi.com/1424-8220/25/2/367>
- [61] Madgwick, S. O. H., Harrison, A. J. L., & Vaidyanathan, R. (2011). Estimation of IMU and MARG orientation using a gradient descent algorithm. *Proc. IEEE International Conference on Rehabilitation Robotics (ICORR 2011)*. DOI 10.1109/ICORR.2011.5975346. Low-cost orientation filter for wearable motion tracking; reported orientation error below 0.8° static and 1.7° dynamic (RMS). **Status: V.**
- [62] Validation studies of inertial sensors (IMUs) for upper-limb angles against optical motion capture: shoulder errors of about 10–20° (RMSE) for one commercial system during simple and complex movements; another system below 4° for most joint angles; accuracy varies with system, joint and calibration. **Status: P.** <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9364332/>
- [63] University of Groningen study: effects of different IMU sensor-to-segment calibration methods (N-pose, functional, manual alignment) on upper-limb kinematics; differences mostly small; the static N-pose recommended as the simplest method for the shoulder. **Status: P.** <https://research.rug.nl/en/publications/effects-of-different-inertial-measurement-unit-sensor-to-segment/>
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- [65] Xsens MVN Link inertial motion-capture suit: US\$12,430 (April 2021 list price); later entry-level estimates US\$7,500–12,000; sold by quotation. **Status: V.** <https://www.cgchannel.com/2021/04/xsens-cuts-price-of-its-entry-level-inertial-mocap-systems/>

P. Handloom, puppetry and craft-capture projects

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- [67] Ministry of Textiles, Rajya Sabha answers: training for handloom workers runs through 28 Weavers' Service Centres; skill upgradation under SAMARTH reached 56,934 weavers from 2015–16 to 2020–21. **Status: P.** https://rsdebate.nic.in/bitstream/123456789/719227/1/PO_254_22072021_U441_p355_p355.pdf
- [68] Textiles Committee. Qualification 'Handloom Weaving – Two Treadles' (TC HLM 06, NSQF Level 3): 315 hours of hands-on training, covering motions of weaving, two-treadle weaving and quality maintenance. **Status: V.** https://textilescommittee.nic.in/sites/default/files/2021-07/Handloom_Weaving_Two_Treadles.pdf
- [69] Siddiqui, L. A., Banerjee, A., Chokhandre, P., & Unisa, S. (2021). Prevalence and predictors of musculoskeletal disorders (MSDs) among weavers of Varanasi, India: a cross-sectional study. *Clinical Epidemiology and Global Health*. n = 364; 12-month prevalence in handloom vs powerloom weavers: upper back 84% vs 45%, lower back 82% vs 50%, shoulder 76% vs 42%. **Status: V.** https://oalib-perpustakaan.upi.edu/Record/doi_18bf152015c74b8aae5783c503ac2069

- [70] Automatic inspection of woven fabric density based on digital image analysis (wavelet-based image method). Relative error below 0.86% compared with manual counting across three basic weaves. **Status: P.** <https://global-sci.com/article/86550/automatic-inspection-of-woven-fabric-density-based-on-digital-image-analysis>
- [71] UNIMA World Encyclopedia of Puppetry Arts, 'Kathputli': strings are looped around the puppeteer's hand and fingers, with no control bar; usually two to five strings (the rasdhari dancer up to nine). Bhat and Nat communities, mostly from Nagaur. **Status: V.** <https://wepa.unima.org/en/?p=3094>
- [72] India's four puppet families: string (Kathputli, Kundhei, Gombeyatta, Bommalattam), shadow (Tholu Bommalata, Togalu Gombeyatta, Ravanachhaya), rod (Putul Nautch, Yampuri) and glove (Pavakoothu of Kerala). **Status: V.** <https://compass.rauias.com/art-culture/puppet-forms-india/>
- [73] People's Archive of Rural India (PARI), 'Delhi's kathputli artists: puppeteers in limbo', and press reports: after the 2017 demolition of the Kathputli Colony at Shadipur, about 2,800 families of puppeteers and other folk artists were to move to a transit camp at Anand Parbat. **Status: V.** <https://ruralindiaonline.org/en/articles/delhis-kathputli-artists-puppeteers-in-limbo>
- [74] CRAEFT: Craft Understanding, Education, Training, and Preservation for Posterity and Prosperity. EU-funded project from 1 March 2023, three years, budget €3,112,106; digital aids, craft simulators and haptics for craft training. **Status: V.** https://www.isti.cnr.it/en/research/project-detail/14501/CRAEFT_Craft_Understanding_Education_Training_and_Preservation_for_Posterity_and_Prosperty
- [75] Mingei (EU Horizon 2020): silk-weaving pilot at the Haus der Seidenkultur, Krefeld; motion capture of craftspeople's gestures and tool use for education, AR/VR and virtual characters. **Status: V.** <https://waag.org/en/project/mingei>

Q. Validation methods

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- [77] Lakens, D. (2017). Equivalence tests: a practical primer for t tests, correlations, and meta-analyses. Social Psychological and Personality Science. Two one-sided tests (TOST) to show that an effect is smaller than a pre-set bound. **Status: V.** <https://pmc.ncbi.nlm.nih.gov/articles/PMC5502906/>

R. Component datasheets and engineering standards

- [78] Bosch Sensortec. BMI270 6-axis IMU datasheet (BST-BMI270-DS000). LGA package 2.5 × 3.0 × 0.83 mm; 16-bit accelerometer and gyroscope; 685 µA typical in full operation; 2 KB FIFO; I2C and SPI. **Status: V.** <https://cdn.sparkfun.com/assets/9/a/2/9/6/bst-bmi270-ds000.pdf>
- [79] Texas Instruments. DRV2605L datasheet (SLOS854): 2–5.2 V supply; VSSOP-10 3.00 × 3.00 mm or DSBGA 1.50 × 1.50 mm; closed-loop LRA auto-resonance; fixed 7-bit I2C address 0x5A (as used in open driver code), so several drivers need an I2C switch. **Status: P.** <https://ti.com/lit/gpn/drv2605l>
- [80] Texas Instruments. TCA9548A low-voltage 8-channel I2C switch (SCPS207): addresses 0x70–0x77, Standard and Fast mode (100 and 400 kHz), one control register enables each channel. **Status: V.** <https://www.ti.com/document-viewer/lit/html/SCPS207H/GUID-170D1E15-60B5-418E-AEF2-45D0CB1F14A2>
- [81] Espressif. ESP32-S3-MINI-1 module datasheet: 15.4 × 20.5 × 2.4 mm; Xtensa LX7 dual core up to 240 MHz; 2.4 GHz Wi-Fi and Bluetooth 5 (LE); PCB antenna. **Status: V.** https://www.espressif.com/sites/default/files/documentation/esp32-s3-mini-1_mini-1u_datasheet_en.pdf
- [82] Espressif. ESP32-S3 series datasheet, current consumption: modem-sleep about 33–47 mA at 80 MHz with both cores; Bluetooth LE transmit about 130 mA (typical). **Status: P.** https://documentation.espressif.com/esp32-s3_datasheet_en.html
- [83] Espressif. ESP-IDF ADC continuous mode driver for ESP32-S3: DMA sampling of one or more channels; default maximum sampling frequency 83,333 Hz. **Status: V.** https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/api-reference/peripherals/adc/adc_continuous.html
- [84] Precision Microdrives. C08-005 8 mm coin linear resonant actuator (LRA): 235 Hz resonance; 8.0 mm diameter × 3.3 mm; 1.8 V rated; 75 mA typical, 102 mA maximum; 1.28 G typical vibration amplitude. **Status: V.** <https://www.precisionmicrodrives.com/?p=12426>

- [85] Microchip. MCP73831 single-cell Li-ion/Li-polymer charge management controller: SOT-23-5 or 2 × 3 mm DFN; charge current programmable from 15 to 500 mA; 4.20 V regulation option ($\pm 0.75\%$); thermal regulation. **Status:** V. <https://www.radiolocman.com/datasheet/data.html?di=430997>
- [86] Murata. 7BB-27-4L0 piezoelectric diaphragm: 27 mm diameter × 0.54 mm; resonant frequency 4.6 kHz; capacitance 20 nF. **Status:** V. <https://nz.element14.com/murata/7bb-27-4/piezo-diaphragm-4-6khz-27mm/dp/2443198>
- [87] NXP Semiconductors. UM10204, I2C-bus specification and user manual: maximum bus capacitance 400 pF per line for Standard and Fast mode (550 pF for Fast-mode Plus); rise time up to 300 ns in Fast mode. **Status:** V. <https://www.nxp.com/docs/en/user-guide/UM10204.pdf>

S. Real-data test, movement identity and instrument sizes

- [88] Gillick, J., Roberts, A., Engel, J., Eck, D., & Bamman, D. (2019). Learning to groove with inverse sequence transformations. Proc. ICML 2019. Groove MIDI Dataset: 13.6 hours, 1,150 MIDI files, over 22,000 measures by 10 drummers on a Roland TD-11 electronic kit, played to a click; drummer, session and style labelled; licence CC BY 4.0. **Status:** V. <https://magenta.tensorflow.org/datasets/groove>
- [89] Liu, J., Zhong, L., Wickramasuriya, J., & Vasudevan, V. (2009). uWave: accelerometer-based personalized gesture recognition and its applications. Pervasive and Mobile Computing, 5(6), 657–675 (first at IEEE PerCom 2009). 98.6% accuracy on over 4,000 samples from 8 users; used for gesture-based user authentication. **Status:** V. https://www.yecl.org/project_uwave.html
- [90] Retail tabla set listings: 5.5-inch dayan (about 10.5 inches tall) with a 9-inch bayan (about 10.25 inches tall). **Status:** V. <https://darbukaplanet.com/products/banjira-standard-tabla-set-aluminum-bayan-and-5-50-dayan>
- [91] Retail listings of handmade Rajasthani Kathputli string puppets: wooden head and torso, cloth body; examples 43–58 cm tall. **Status:** V. <https://itokri.com/products/564547-30-rajasthani-men-handmade-puppet-kathputli>

T. Research methods

- [92] Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Erlbaum. **Status:** S.

Appendix A. Partner kit for three crafts

Craft	Where to find masters
Tabla	Gurus empanelled with a Zonal Cultural Centre under Guru-Shishya Parampara [28]; tabla faculty at music colleges; Gandharva Mahavidyalaya centres [31]
Handloom	Weavers' Service Centres [67]; master weavers and trainers of the two-treadle qualification [68]
Puppetry	Kathputli families and troupes, including the former Kathputli Colony community [73]; puppeteers known to Zonal Cultural Centres and Sangeet Natak Akademi [27]

Outreach message

Respected Guruji / Ustad, we are engineering students building PARAMPARA for Smart India Hackathon 2026. It records how a master moves while playing, weaving or working a puppet (light sensors on the back of the hand and arm, nothing on the fingertips or palm), so that learners can practise against it, with the master's consent and credit. We would be grateful for two short sessions of about 45 minutes on different days. You decide how the recording may be used, you will receive a copy, and you can withdraw it at any time.

Pranam Guruji / Ustad ji, hum engineering students hain aur Smart India Hackathon 2026 ke liye PARAMPARA bana rahe hain. Isme guru ke haath aur baanh ki gati record hoti hai (haath ke peeche aur baanh par halke sensor, ungliyon ke siron aur hatheli par kuch nahi), taaki shishya uske saath abhyas kar sakein, guru ki anumati aur naam ke saath. Kya aap alag-alag din do baar lagbhag 45 minute de sakte hain? Recording ka upyog aap tay karenge, aapko copy milegi, aur aap kabhi bhi wapas le sakte hain.

Consent: I agree that...	Yes / No
my movements may be recorded with sensors on the back of my hands and arms, and a sensor on my instrument, loom or puppet	☐ / ☐
I may be asked to perform on a second instrument, loom or puppet	☐ / ☐
the recording may be used to teach learners through PARAMPARA	☐ / ☐
my name, gharana or community may be shown with the recording	☐ / ☐
the recording may be shown at Smart India Hackathon 2026	☐ / ☐
the recording may be used commercially (only with a separate written agreement)	☐ / ☐
I understand I can withdraw at any time, and I will receive a copy of my recording	☐

Signature, date and witness; a spoken consent recording in the master's language is kept with the form. The master receives a copy of the recording and fingerprint, credit wherever it is used, control (approve, limit, withdraw) and a share of any licence income as agreed in writing.

Appendix B. Experiment E1 protocol

	Day 1	Day 2 (sleeve re-fitted)	Cycles
Master A	Instrument 1, then 2	Instrument 2, then 1	4 × 20 = 80

	Day 1	Day 2 (sleeve re-fitted)	Cycles
Master B	Instrument 2, then 1	Instrument 1, then 2	80
Master C	Instrument 1, then 2	Instrument 2, then 1	80

1. **Set-up:** sleeves on both arms, 20-second N-pose; tool sensor fitted; tuning, loom settings or puppet noted.
2. **Recording:** at least 20 uninterrupted cycles per instrument: Teentaal theka at 80 beats per minute (tabla); 20 picks of plain weave (handloom); a fixed gesture phrase repeated 20 times (puppetry).
3. **E0 block:** on day 1, alternating blocks of 10 cycles with and without the sleeve on instrument 1.
4. **Export** one row per cycle: master, session, instrument, f1 ... fn.
5. **Analyse:** python3 PARAMPARA/sim/skilltwin_validation.py analyze e1.csv; report T-a, T-b, T-c, the control and the verdict.

Result template for the deck

Craft ____ · masters A, B, C · new day ____% · new instrument ____% (chance 33%) · confound gap ____,
p = ____ · instrument control ____% (chance 50%) · claim (Section 8.1) ____.

Appendix C. Protocols for E0 and E2–E5

#	Procedure	Measure	Pass [Target]
E0	Blocks: no sleeve, sleeve, no sleeve, sleeve; 10 cycles each; tool sensor on throughout; comfort rating	Differences in timing, cycle time and strength; two one-sided tests [77]	Within ± 10 ms, $\pm 5\%$, ± 1 dB; comfort ≥ 4 of 5
E2	Weakest felt cue (up-down method) at rest and while performing slowly; finger, wrist, elbow and shoulder; cue before vs on the beat	Ratio of performing to rest strength	≤ 2 at one or more sites per joint
E3	Feel and hear A, B, then X (random). 20 rounds	Correct out of 20	≥ 15 ($p = 0.02$)
E4	40 random cues across 5 fingers and 3 joints at 2 strengths while performing	Percentage correct	$\geq 90\%$ sites, $\geq 80\%$ strengths
E5	16 beginners randomised 8 and 8: video practice or PARAMPARA; three 20-minute sessions on three days; retention 48 h later, device off, blinded tester	Unaided score at retention	Difference and 95% CI; full trial if PARAMPARA $\geq$ video

Appendix D. Demo day checklist

- Two sleeves plus one spare; rings in three sizes; spare cells, power bank, cables; microSD cards.
- Tabla and practice pad; a Kathputli with its pod; handloom recordings and a cloth sample.
- Pre-calibrated judge profile with a 20-second N-pose re-check; offline app copy.

- Masters' fingerprints and E1 verdicts; one signed recording and one edited copy.
- Printed one-page summary, consent summaries, bench-test sheet (H1–H8).
- Backup video; three rehearsals completed without help.

Appendix E. Simulations (planning tools, not evidence of learning)

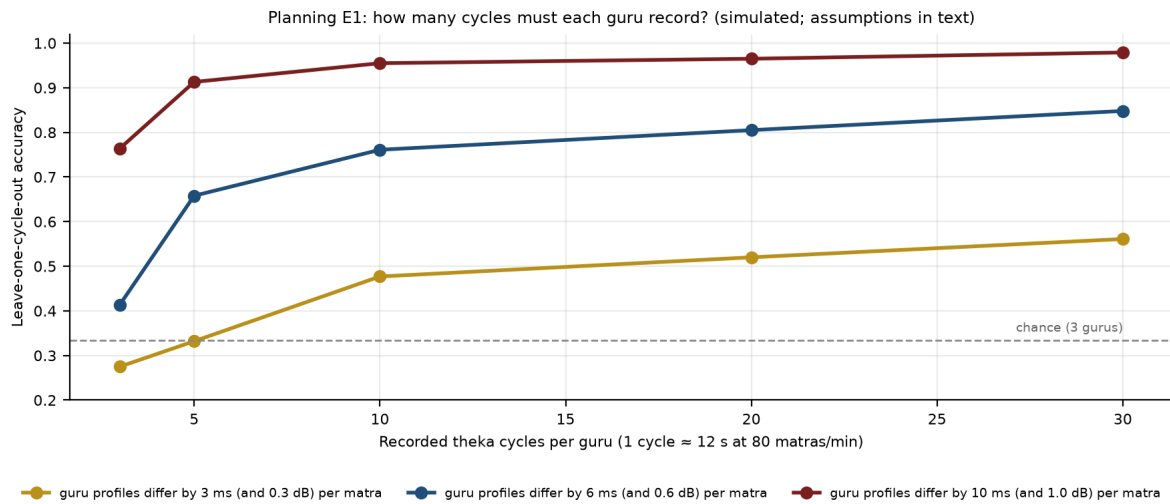


Figure E1. With assumed variation between tabla cycles (15 ms, 1.5 dB) and 20 cycles per player, profiles differing by about 6 ms and 0.6 dB per beat are told apart about 80% of the time.

Unaided score on day 4 (simulated)	Assumption A	Assumption B	Assumption C
Cues always on	0.48	0.42	0.57
No cues	0.46	0.58	0.71
Fixed fade	0.67	0.63	0.68
Check-cycle fade (PARAMPARA)	0.69	0.65	0.68

Cues that never stop do poorly under every assumption; check-cycle fading did slightly better than a fixed fade under two of three (+0.015, 95% CI 0.013–0.017). People, not simulations, decide (E5 and the full trial). On synthetic data the confound test passes a master-driven world and fails an instrument-driven one (Figure 11).

Appendix F. Reference check and parameters

Original citation in the team's first documents	Finding
'Masur & Sacks, Gesture-Based Adaptive Haptic Guidance, IEEE Robotics (2025)'	Does not exist; closest real work is Zahedi et al. 2017 [11]
'Flandorfer et al., Wearable Haptic Learning Systems, Sensors (2022)'	Not found; replaced by a verified review [10]
'Grindlay (2008), ACM Multimedia'	Wrong venue; correct sources cited [1, 2]

Parameter	Value
IMU sampling	200 samples per second (FIFO burst reads)
Tabla piezo sampling	16,000 samples per second per channel (DMA)
I2C buses	2 per sleeve, 400 kHz, TCA9548A on each
Calibration	20-second N-pose each session
Cue length	30–80 ms
Cue strength	$\geq 3 \times$ the person's weakest felt level, firmware maximum
Check cycles	session start + every 4 cycles
Target to move on	unaided score $\geq 85\%$ on 2 days + master's approval
E1 pass	T-a and T-b $\geq 70\%$; T-c $p < 0.01$
Fingerprint recording	≥ 20 cycles per master, day and instrument

Appendix G. CAD and code index

File	What it is
PARAMPARA/cad/parampara_cad.py	Parametric CAD of every housing, the sleeve assembly and the three tool kits; exports STEP and STL
PARAMPARA/cad/out/.step, .stl	Ring, motor, joint, hand-board and hub housings and lids, ready to inspect or print
PARAMPARA/cad/render_all.py, compose_figures.py	Off-screen renders and the labelled figures in Section 4
PARAMPARA/sim/gmd_fingerprint.py, make_gmd_figure.py	Real-data test on the Groove MIDI Dataset (Section 7)
PARAMPARA/sim/skilltwin_validation.py	E1 confound test (T-a, T-b, T-c, control, verdict)
PARAMPARA/sim/fade_engine_sim_v2.py, fingerprint.py	Fade simulation and E1 recording-length planning
PARAMPARA/lite/index.html, e2e_test.py	Phone prototype (open on Android Chrome for vibration) and its end-to-end browser test
PARAMPARA/firmware/src/*.h, test/test_main.cpp	Sleeve firmware core (drivers, address map, fade, scoring, clock sync) and 27 host unit tests

```
python3 PARAMPARA/cad/parampara_cad.py          # STEP/STL files
python3 PARAMPARA/cad/render_all.py && python3 PARAMPARA/cad/compose_figures.py
python3 PARAMPARA/sim/gmd_fingerprint.py groove/ # real-data test
make -C PARAMPARA/firmware test                  # 27 firmware unit tests
python3 PARAMPARA/lite/e2e_test.py                # phone prototype end-to-end test
node PARAMPARA/build/build_v6.js                  # this document
```